Name ____ Date ___________ Period _____

Solving Absolute Value Linear Equations

TEACHER KEY | Algebra II Honors | Unit 2 | Day 5 | TEK A2.6.E

Objective: I can justify why an absolute value equation splits into two cases, and solve absolute value linear equations that require multiple isolation steps.

Vocabulary

Isolate — get the absolute value expression alone before splitting.

Split into two cases — $|A| = k$ $(k>0)$ becomes $A = k$ OR $A = -k$.

Part 1 — WHY Does |A| = k Split Into Two Cases?

Using Day 1's piecewise definition of absolute value, explain why $|A| = k$ $(k > 0)$ means $A = k$ OR $A = -k$.

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Sample: **if $A \geq 0$, $|A| = A$, so $A = k$ directly. If $A < 0$, $|A| = -A$, so $-A = k$, which means $A = -k$. Both cases are needed to cover every possible sign of A**

Part 2 — Isolate First, Then Split

Example 1. $2|x + 1| - 3 = 7$

Isolate: $|x + 1| =$ **5** Solve: $x =$ **4** or $x =$ **-6**

Example 2. $5 - 2|x - 1| = -3$

Isolate: $-2|x - 1| =$ **-8** → $|x - 1| =$ **4** Solve: $x =$ **5** or $x =$ **-3**

Notice: dividing by a NEGATIVE number does not flip anything here, unlike an inequality - it just isolates the absolute value.

Part 3 — Fractions and Distributing

Example 3. $|(2x - 4)/3| = 6$

$2x - 4 = 18$ or $2x - 4 =$ **-18** $x =$ **11** or $x =$ **-7**

Example 4. $|3(x - 2)| + 1 = 10$

Isolate: $|3(x - 2)| =$ **9** → $3(x - 2) = 9$ or $3(x - 2) =$ **-9** $x =$ **5** or $x =$ **-1**

You Try. $|4(x + 1)| - 2 = 14$

$|4(x+1)| =$ **16** $x =$ **3** or $x =$ **-5**

Practice

1. $4|x + 2| = 20$ $x =$ **3** or $x =$ **-7**

2. $|(x + 5)/2| = 3$ $x =$ **1** or $x =$ **-11**

3. $8 - 3|x - 2| = -1$ $x =$ **5** or $x =$ **-1**

Exit Ticket

Solve $6 - 2|x + 3| = -4$, and explain in one sentence why the equation splits into two cases once $|x+3|$ is isolated.

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$|x + 3| =$ **5** $x =$ **2** or $x =$ **-8** **x+3 could be positive or negative and still have absolute value 5**